

P versus NP: Where the Adaptive Template Breaks

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Abstract

We apply the adaptive regularization template — regularize, bound uniformly, extract by compactness, inherit regularity — to the P vs NP problem and identify precisely where it breaks. For the other five Clay problems the template identifies a single named open step; for P vs NP, the uniform bound step fails structurally. The failure is equivalent to $\mathbf{P} \neq \mathbf{NP}$: a uniform polynomial bound on the search complexity of an NP-complete problem is exactly a polynomial-time algorithm for that problem. We show the template breakdown is consistent with the three known proof barriers (relativization, natural proofs, algebrization). The breakdown is itself a structural observation about the problem.

1 The Template and Its Applications

The adaptive regularization template:

1. **Regularize.** Introduce parameter $\alpha > 0$ penalizing pathological behavior; the regularized problem is solvable for each $\alpha > 0$.
2. **Bound.** Establish a uniform estimate independent of α .
3. **Compact.** Extract a convergent subsequence as $\alpha \rightarrow 0^+$.
4. **Regularity.** The limit inherits the desired property.

In the other five Clay problems: Navier-Stokes closes at large data (Step 2 open for large data, closes for small); Yang-Mills has two conditional steps (lattice gap, continuum limit); Hodge reduces to the Complex Regularity Conjecture (Step 4); Riemann reduces to the Hilbert-Pólya conjecture (Step 2, operator construction); BSD closes for rank 0–1 and is conditional on GGZ_r for rank ≥ 2 . In each case, the template terminates at a named open conjecture.

For P vs NP, the template terminates differently: the uniform bound *cannot exist* without solving the problem itself.

2 Setup

A language $L \subseteq \{0, 1\}^*$ is in **NP** if there is a polynomial-time verifier and a polynomial p such that $x \in L \iff \exists w, |w| \leq p(|x|), V(x, w) = 1$. **P** is the class of languages decidable in deterministic polynomial time.

P vs NP. Is $\mathbf{P} = \mathbf{NP}$?

For a language L and $n \in \mathbb{N}$, let $T^*(L, n)$ denote the optimal deterministic time to decide all inputs of length n :

$$T^*(L, n) = \min_{A \text{ decides } L} \max_{|x|=n} T_A(x).$$

Definition 1 (Search hardness). The *search hardness* of L at length n is

$$\psi(L, n) = \frac{T^*(L, n)}{n^k}$$

where k is the smallest integer such that $T^*(L, n) = O(n^k)$ (if such k exists; otherwise $\psi(L, n) \rightarrow \infty$).

The regularized complexity is

$$C_\alpha(L, n) = T^*(L, n) + \alpha \cdot \psi(L, n)^2.$$

For $L \in \mathbf{P}$: $\psi(L, n) = O(1)$, so $C_\alpha(L, n) = O(1)$ uniformly in α . For L NP-complete (assuming $\mathbf{P} \neq \mathbf{NP}$): $\psi(L, n) \rightarrow \infty$, and C_α grows with α as $n \rightarrow \infty$.

3 Where the Template Breaks

Proposition 2 (Step 2 fails: no uniform bound). *For any NP-complete language L , the following are equivalent:*

- (a) $\psi(L, n) \leq C$ for some constant C and all n ;
- (b) $L \in \mathbf{P}$.

Proof. (b) $\Rightarrow$ (a): If $L \in \mathbf{P}$ then $T^*(L, n) = O(n^k)$ for some fixed k , so $\psi(L, n) = T^*(L, n)/n^k = O(1)$.

(a) $\Rightarrow$ (b): A uniform bound $\psi(L, n) \leq C$ means $T^*(L, n) \leq C \cdot n^k$ for all n , i.e., L is decided in time $O(n^k)$, so $L \in \mathbf{P}$. Since L is NP-complete, $\mathbf{P} = \mathbf{NP}$. $\square$

Proposition 3 (Step 3 fails: no compactness). *Assuming $\mathbf{P} \neq \mathbf{NP}$, the family of algorithms $\{A_\alpha\}_{\alpha>0}$ minimizing $C_\alpha(L, n)$ has no polynomial-time limit as $\alpha \rightarrow 0^+$.*

Proof. Any polynomial-time limit A_0 would decide L in polynomial time, giving $L \in \mathbf{P}$, contradicting $\mathbf{P} \neq \mathbf{NP}$. $\square$

Remark 4. The contrast with Navier-Stokes is sharp. In NS, the uniform H^1 bound fails for large data but is not *equivalent* to regularity — it is a sufficient condition. Here, the uniform polynomial bound on ψ is *exactly* the statement $L \in \mathbf{P}$: necessary and sufficient. There is no weaker bound that would do.

4 Consistency with the Three Barriers

Relativization [2]. There exist oracles $\mathcal{O}$ with $\mathbf{P}^{\mathcal{O}} = \mathbf{NP}^{\mathcal{O}}$ and oracles $\mathcal{O}'$ with $\mathbf{P}^{\mathcal{O}'} \neq \mathbf{NP}^{\mathcal{O}'}$. The search hardness $\psi(L, n)$ is defined via oracle running times; thus ψ and C_α relativize. The template’s uniform bound would hold relative to $\mathcal{O}$ and fail relative to $\mathcal{O}'$. A proof via the template would relativize — but Baker-Gill-Solovay shows no relativizing proof can separate $\mathbf{P}$ from $\mathbf{NP}$.

Natural proofs [3]. Razborov-Rudich: any proof technique that is *constructive* (produces a distinguishing property) and *large* (applies to most functions) cannot separate $\mathbf{P}$ from $\mathbf{NP}$ under standard cryptographic assumptions. The uniform bound argument in the template establishes a constructive property of the complexity function and is large (applies at every input length). It is therefore a natural proof technique and falls within the Razborov-Rudich barrier.

Algebrization [4]. Aaronson-Wigderson: algebraic extensions of relativization (low-degree extension queries) block most algebraic techniques. The compactness step in the template uses limiting arguments that can be algebraized; the barrier applies.

Remark 5 (Why barriers don’t block the other five problems). For NS, YM, Hodge, RH, BSD: the template operates on analytic or geometric objects (PDEs, currents, operators, L -functions), not on Boolean circuits. The barriers are complexity-theoretic and do not apply. This is why the template terminates at named analytic conjectures for the other five problems, rather than at a barrier.

5 The Structural Observation

Theorem 6. *The adaptive regularization template applied to P vs NP :*

1. *Terminates at Step 2 with a failure equivalent to $\mathbf{P} \neq \mathbf{NP}$ (Proposition 2).*
2. *Cannot be rescued by a non-relativizing, non-natural, non-algebrizing technique using the template’s structure, since the template itself relativizes, is natural, and algebrizes.*

This is consistent with — and predicted by — the three known barriers.

Remark 7 (No computational viscosity). In Navier-Stokes, the $\alpha|\nabla u|^2$ term activates when velocity gradients concentrate, providing dissipation that prevents blowup. P vs NP has no analogue: there is no computational mechanism that activates when search hardness concentrates and prevents NP -hard instances from forming. Hardness in computational complexity is not self-regulating. This is the sense in which P vs NP is structurally different from the other five Clay problems.

References

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